Virology from the Perspective of Soft Condensed Matter Physics

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I. Soft Condensed Matter and Life Science

Soft condensed matter physics studies materials that lie between simple liquids and crystalline solids. Such materials combine solid-like elasticity with liquid-like flow, with typical examples including polymers, colloids, surfactants, liquid crystals, gels, and foams. A defining feature of soft matter is that its characteristic interaction energies are often comparable to the thermal energy $k_B T$, so its behaviour is strongly shaped by intermolecular forces and thermal fluctuations, typically on nano- to micrometre length scales [1].

From this perspective, much of biology can be viewed as soft matter. DNA, RNA, proteins, membranes, ion channels, the cytoskeleton, and even whole cells all belong to this broad physical domain. Biological function is therefore not determined by chemistry alone, but also by mechanical response, fluctuation, self-organisation, and transport in soft and complex environments [1, 2]. In this sense, life itself can be understood as a soft machine. Soft condensed matter physics thus provides a useful framework for understanding biological processes.

II. Viruses as Soft Condensed Matter

A virus is, in essence, a delivery system for genetic material. Its basic task is to introduce a DNA or RNA genome into a host cell, where the host machinery can then express and replicate that genome. Structurally, viruses are simple: they contain genetic material enclosed by a protein shell called a capsid. Some viruses also possess a lipid envelope derived from the host cell membrane, as in HIV, influenza virus, and SARS-CoV-2 [3, 4]. Since viruses are built from proteins, nucleic acids, and, in many cases, lipids, they are composed entirely of materials that fall naturally within the domain of soft matter [1, 8].

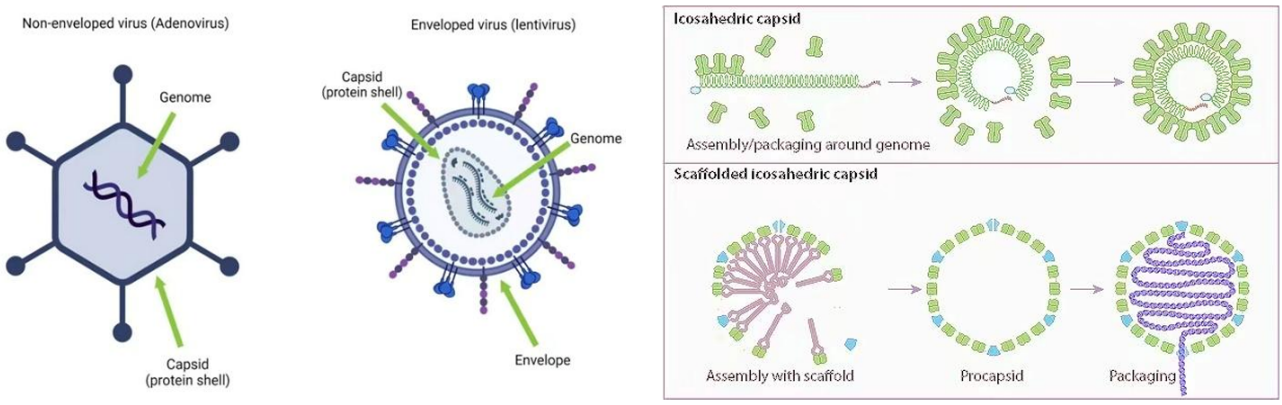

Figure 1: Structural comparison between non-enveloped and enveloped viruses, together with two schematic routes of viral assembly. Non-enveloped viruses are enclosed by a protein capsid alone, whereas enveloped viruses additionally possess a host-derived lipid membrane. Viral particles may assemble either through capsid formation around the genome or through capsid formation followed by genome packaging [3, 4].

More importantly, viruses do not merely contain soft matter; they also behave like soft matter systems. They are highly ordered supramolecular assemblies whose physical properties depend on a balance between stability and flexibility [4]. On

the one hand, viral capsids must be sufficiently robust to protect the genome and tolerate external stresses. On the other hand, they must also remain responsive to environmental change so that genome release can occur at the appropriate stage of infection. Their mechanical behaviour reflects this compromise: viral particles are neither rigid like conventional solids nor disordered like simple fluids, but dynamically organised structures with both resilience and adaptability [5].

This perspective becomes even clearer when one considers viral self-assembly. After infection, the host cell produces viral proteins and genomes, which then organise into new viral particles. In broad terms, this can occur in two ways: capsid proteins may assemble directly around the genome, or an empty capsid may form first and later package the genome [3, 4]. In both cases, the process relies on selective but reversible interactions among proteins, nucleic acids, and the surrounding medium. This is a characteristic theme in soft matter physics, where weak interactions, Brownian motion, and collective organisation allow complex structures to emerge spontaneously. Viruses are therefore not only made of soft matter; they are also shaped by the physical principles of soft matter.

III. A Real Case: Soft Matter Physics and the COVID-19 Pandemic

The relevance of soft condensed matter physics became especially visible during the COVID-19 pandemic. In a 2020 perspective article, SARS-CoV-2 was described as a "quintessential soft matter object": a composite colloid made up of surfactant-like and polymeric components [6]. Within this framework, the viral lipid envelope can be treated as a surfactant-rich interface, while the spike proteins and RNA contribute polymer-like structural and dynamical features. This description connects virology with familiar soft matter processes such as diffusion, evaporation, adhesion, and transport in complex fluids [6, 7].

Such a viewpoint is not merely conceptual; it also helps explain everyday public health practice. The viral envelope can be disrupted by surfactants such as soap, which is why cleaning and disinfection are effective in reducing viral infectivity [6, 8]. Likewise, masks and related protective materials function by filtering virus-laden droplets through porous fibrous networks, where fluid flow, particle capture, and interfacial interactions all matter [6]. In this sense, soft matter physics contributed not only to a better scientific understanding of SARS-CoV-2, but also to practical strategies of protection, filtration, and transmission control.

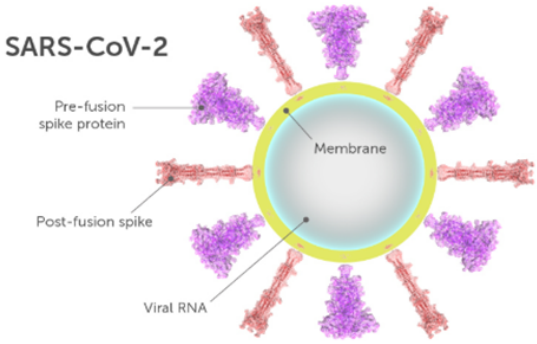

Figure 2: Coronavirus spike proteins embedded in the viral envelope. For enveloped viruses such as SARS-CoV-2, the spike-decorated lipid membrane is central to how the particle interacts with droplets, surfaces, and host cells, making it a natural focus for a soft matter description [6, 9].

Overall, virology and soft condensed matter physics meet at a productive interface. Viruses are biological entities, but they are also physical assemblies whose structure, mechanics, and transport can be analysed using the language of soft matter. This makes soft condensed matter physics a useful and increasingly important perspective in modern virology.

References

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